

LEWISTOWN, MT, USA

WMO: 726776

Lat: 47.049N

Lon: 109.458W

Elev: 1263

StdP: 87.04

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24036

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.9	(c) -22.4	(d) -29.8	(e) 0.3	(f) -25.0	(g) -25.8	(h) 0.4	(i) -21.9	(j) 14.7	(k) 4.6	(l) 13.1	(m) 4.1	(n) 2.9	(o) 130	(p) 0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.0	(c) 32.3	(d) 15.9	(e) 30.5	(f) 15.6	(g) 28.7	(h) 15.1	(i) 18.0	(j) 27.9	(k) 16.9	(l) 26.9	(m) 16.0	(n) 26.1	(o) 4.0	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	12.0	21.1	13.3	11.1	20.0	12.3	10.4	19.2	55.5	27.8	52.1	26.7	49.2	26.1	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.3	(b) 10.9	(c) 9.2	(e) -30.6	(f) 35.4	(g) 3.9	(h) 1.5	(i) -33.4	(j) 36.5	(k) -35.6	(l) 37.3	(m) -37.8	(n) 38.2	(o) -40.7	(p) 39.2		
(a) 12.3	(b) 10.9	(c) 9.2	(e) -30.2	(f) 19.7	(g) 3.3	(h) 1.1	(i) -32.6	(j) 20.5	(k) -34.6	(l) 21.2	(m) -36.4	(n) 21.8	(o) -38.8	(p) 22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-4.5	-4.4	0.4	4.9	9.8	14.4	19.4	18.6	13.4	6.5	0.4	-4.3
	DBStd	10.32	7.76	7.81	6.87	5.19	4.40	3.86	3.45	3.52	4.91	5.58	7.11	7.41
	HDD10.0	2273	449	403	300	165	58	7	0	1	22	133	292	443
	HDD18.3	4533	707	636	556	403	266	128	31	41	159	367	539	702
	CDD10.0	917	0	1	2	12	52	139	290	267	124	25	4	0
	CDD18.3	133	0	0	0	0	1	9	62	49	11	1	0	0
	CDH23.3	2377	0	0	0	5	39	184	996	857	276	19	0	0
	CDH26.7	876	0	0	0	0	4	48	402	329	89	3	0	0
Wind	WSAvg	4.1	4.6	4.3	4.4	4.3	4.1	3.8	3.3	3.4	3.6	4.2	4.6	4.6
Precipitation	PrecAvg	456	25	17	24	30	84	78	45	48	39	28	18	22
	PrecMax	714	70	50	66	68	233	164	123	141	98	62	38	60
	PrecMin	285	2	2	5	9	21	25	3	7	2	0	2	4
	PrecStd	117	17	9	15	17	50	40	30	37	27	17	11	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	14.0	19.3	23.8	27.0	31.1	34.9	34.6	31.9	26.0	19.8	12.4
		MCWB	4.7	5.6	7.9	10.6	13.8	16.9	17.1	16.2	14.9	11.8	8.9	5.1
	2%	DB	9.1	10.7	15.9	20.5	24.1	27.8	32.6	32.2	29.3	22.1	15.9	9.3
		MCWB	3.2	4.2	6.8	9.5	12.8	16.2	16.4	15.4	13.9	10.2	7.0	3.5
	5%	DB	7.0	7.8	12.6	17.5	21.8	25.3	30.9	30.2	26.8	18.8	12.8	7.2
		MCWB	2.3	2.6	5.3	8.3	11.9	15.2	16.2	15.2	13.1	9.2	5.5	2.3
	10%	DB	5.2	5.4	9.8	14.3	19.0	23.0	28.9	28.3	23.7	15.8	10.2	5.3
		MCWB	1.4	1.3	3.9	6.6	10.5	14.1	15.7	14.7	12.2	7.7	4.3	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	6.4	8.7	11.8	15.2	19.0	19.6	18.0	16.6	12.9	9.7	5.7
		MCDB	11.4	13.1	17.8	21.7	23.7	27.6	29.5	28.9	26.5	23.7	18.9	11.5
	2%	WB	3.7	4.5	7.2	10.0	13.7	17.3	18.4	16.9	15.1	10.9	7.4	3.9
		MCDB	8.5	10.1	15.2	19.1	22.1	25.5	28.7	28.1	26.6	19.9	14.9	8.6
	5%	WB	2.5	2.9	5.6	8.6	12.5	15.9	17.4	16.0	13.9	9.6	5.9	2.6
		MCDB	6.5	7.5	12.1	16.8	20.2	23.8	27.4	27.1	24.7	17.3	12.1	6.6
	10%	WB	1.5	1.5	4.1	7.2	11.1	14.7	16.5	15.3	12.6	8.3	4.5	1.5
		MCDB	5.0	5.3	9.4	13.9	17.9	21.9	26.6	26.0	22.3	15.3	9.9	5.1
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.5	11.4	12.2	12.5	13.1	16.0	16.1	14.7	12.7	11.2	10.5
		MCDBR	11.6	12.4	15.3	17.8	17.4	17.4	19.1	19.5	19.7	18.0	15.1	11.6
		MCWBR	7.6	7.9	8.5	9.2	8.6	8.3	7.3	7.4	8.2	8.9	8.5	7.5
		MCDBR	10.8	11.8	14.6	16.5	15.7	15.7	16.7	16.9	17.7	16.3	14.2	11.0
	5% WB	MCWBR	7.4	7.7	8.4	8.8	8.1	8.0	7.5	6.9	7.8	8.6	8.3	7.5
Clear-Sky Solar Irradiance	taub	0.245	0.251	0.266	0.304	0.322	0.326	0.332	0.355	0.310	0.281	0.252	0.241	
	taud	2.524	2.545	2.577	2.462	2.465	2.493	2.495	2.389	2.506	2.572	2.567	2.518	
	Ebn at Noon	881	947	975	955	942	937	925	889	908	893	866	844	
	Edh at Noon	60	73	83	103	106	104	103	110	88	71	57	54	
All-Sky Solar Radiation	RadAvg	1.51	2.48	3.67	4.79	5.71	6.64	7.13	5.94	4.35	2.77	1.71	1.19	
	RadStd	0.10	0.18	0.24	0.24	0.53	0.43	0.33	0.34	0.32	0.20	0.14	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page